SEMESTER 2 EXAMINATION 2020 - 2021

COMPUTATIONAL BIOLOGY

DURATION 120 MINS (2 Hours)

This paper contains 4 questions

Answer all questions. Section A (worth 40 marks) is a series of questions with short answers. Section B (worth 60 marks) involves longer questions

An outline marking scheme is shown in brackets to the right of each question.

This examination is worth 70%. The tutorials was worth 30%.

University approved calculators MAY be used.

A foreign language dictionary is permitted ONLY IF it is a paper version of a direct 'Word to Word' translation dictionary AND it contains no notes, additions or annotations.

9 page examination paper.

Answers

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Section A

Question A1.

(a) For Hidden Markov Models (HMMs) describe why pseudo-counts are commonly used in training the parameters.

Indicative Solution for Question A1(a).

The emission and transition probabilities are computed from the training data, but the training data might be incomplete. This could lead to zero probabilities where in fact there is a small probability of an event happening. Adding pseudo-counts prevents any probabilities being zero compensating for using a finite training set.

[5 marks]

(b) Describe a good way of encoding different amino acids in an artificial neural network.

Indicative Solution for Question A1(b).

Using a one-hot encoding prevents any biases in encoding amino acids. This can be combined with an embedding that may be learnt from a different data set.

[5 marks]

(c) Describe what is meant by coalescence in population genetics.

Indicative Solution for Question A1(c).

Coalescence occurs when we trace the evolutionary history backwards through the generations. After a certain time different genes across the current population have a common ancestor. In asexual reproduction (e.g. in mitrochondrial DNA) coalescence will happen relatively rapidly.

[5 marks]

(d) What do we mean by conditional independence and why is it important in inferring interactions in a gene regulation network from correlation data?

Indicative Solution for Question A1(d).

Conditional independence implies that for two random variables X and Y when conditioned on a third random variable Z then

$$\mathbb{P}(X, Y|Z) = \mathbb{P}(X|Z)\ \mathbb{P}(Y|Z),$$

this doesn't mean we have full independence $\mathbb{P}(X, Y) = \mathbb{P}(X)\ \mathbb{P}(Y)$. Conditional independence suggests a lack of a direct causal link between X and Y despite the fact that they be correlated (if they are not fully independent).

Answers

[5 marks]

(e) Describe how microarrays are used to measure the concentration of RNA.

Indicative Solution for Question A1(e).

Microarrays usually consist of a large number of micro-wells each containing multiple stands of complementary DNA or RNA. Each well would be designed to detect a different RNA. The cells under investigation would have their RNA tagged with a fluorescent dye while control cells would have their RNA tagged with a different coloured fluorescent dye. By measuring the relative activation of the two sets of cells we can determine the relative abundance of RNA strands.

[5 marks]

(f) Explain why we model the time between reaction by an exponential distribution.

Indicative Solution for Question A1(f).

The exponential distribution arises from the assumption that in any infinitesimal time period the probability of an event is independent of any other event and therefore Poisson distributed. The probability of having no events in a succession of periods leads to an exponential distribution.

Answers

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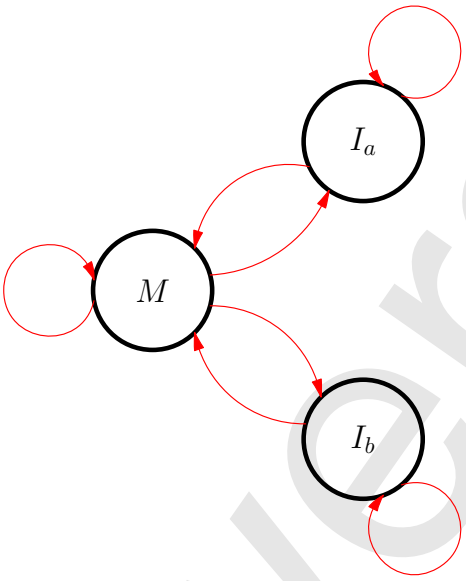
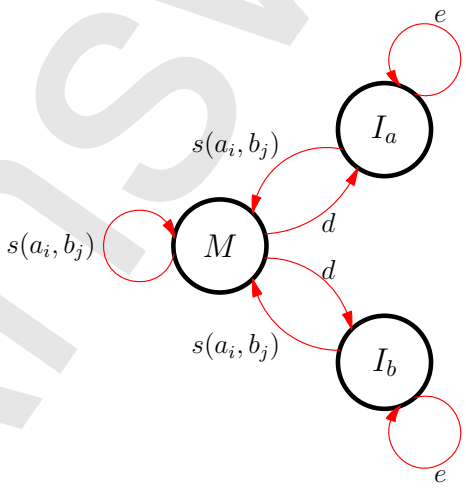
[5 marks]

(g) Below is the finite state automaton modelling affine gaps. Label the transition costs and explain what the three states represent.



Indicative Solution for Question A1(g).



M **represents the case where there is no gap and the amino acids are being substituted**

I_a **corresponds to a gap where we insert into string** a

I_b **corresponds to a gap where we insert into string** b

[10 marks]

Answers

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Section B

Question B1.

(a) Write down all the suffixes of the string "**GTAACGT$**".

Indicative Solution for Question B1(a).

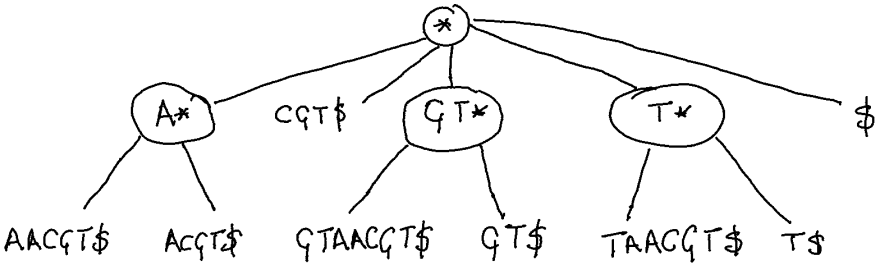
{$, T$, GT$, CGT$, ACGT$, AACGT$, TAACGT$, GTAACGT$}

[5 marks]

(b) Sketch the suffix tree (trie of all suffixes) for "**GTAACGT$**".

Indicative Solution for Question B1(b).

{$, T$, GT$, CGT$, ACGT$, AACGT$, TAACGT$, GTAACGT$}



[10 marks]

(c) How many comparisons are necessary to find a match to the string "**ACG**" in "**GTAACGT$**" using the suffix tree.

Indicative Solution for Question B1(c).

Two comparisons are necessary to reach a leaf.

[5 marks]

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Question B2.

Using the Blosum50 substitution matrix

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	**5**	-2	-1	-2	-1	-1	-1	0	-2	-1	-2	-1	-1	-3	-1	1	0	-3	-2	0
R	-2	**7**	-1	-2	-4	1	0	-3	0	-4	-3	3	-2	-3	-3	-1	-1	-3	-1	-3
N	-1	-1	**7**	2	-2	0	0	0	1	-3	-4	0	-2	-4	-2	1	0	-4	-2	-3
D	-2	-2	2	**8**	-4	0	2	-1	-1	-4	-4	-1	-4	-5	-1	0	-1	-5	-3	-4
C	-1	-4	-2	-4	**13**	-3	-3	-3	-3	-2	-2	-3	-2	-2	-4	-1	-1	-5	-3	-1
Q	-1	1	0	0	-3	**7**	2	-2	1	-3	-2	2	0	-4	-1	0	-1	-1	-1	-3
E	-1	0	0	2	-3	2	**6**	-3	0	-4	-3	1	-2	-3	-1	-1	-1	-3	-2	-3
G	0	-3	0	-1	-3	-2	-3	**8**	-2	-4	-4	-2	-3	-4	-2	0	-2	-3	-3	-4
H	-2	0	1	-1	-3	1	0	-2	**10**	-4	-3	0	-1	-1	-2	-1	-2	-3	2	-4
I	-1	-4	-3	-4	-2	-3	-4	-4	-4	**5**	2	-3	2	0	-3	-3	-1	-3	-1	4
L	-2	-3	-4	-4	-2	-2	-3	-4	-3	2	**5**	-3	3	1	-4	-3	-1	-2	-1	1
K	-1	3	0	-1	-3	2	1	-2	0	-3	-3	**6**	-2	-4	-1	0	-1	-3	-2	-3
M	-1	-2	-2	-4	-2	0	-2	-3	-1	2	3	-2	**7**	0	-3	-2	-1	-1	0	1
F	-3	-3	-4	-5	-2	-4	-3	-4	-1	0	1	-4	0	**8**	-4	-3	-2	1	4	-1
P	-1	-3	-2	-1	-4	-1	-1	-2	-2	-3	-4	-1	-3	-4	**10**	-1	-1	-4	-3	-3
S	1	-1	1	0	-1	0	-1	0	-1	-3	-3	0	-2	-3	-1	**5**	2	-4	-2	-2
T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	2	**5**	-3	-2	0
W	-3	-3	-4	-5	-5	-1	-3	-3	-3	-3	-2	-3	-1	1	-4	-4	-3	**15**	2	-3
Y	-2	-1	-2	-3	-3	-1	-2	-3	2	-1	-1	-2	0	4	-3	-2	-2	2	**8**	-1
V	0	-3	-3	-4	-1	-3	-3	-4	-4	4	1	-3	1	-1	-3	-2	0	-3	-1	**5**

(a) Complete the following table for Needleman-Wunsch algorithm using a cost of $d = 8$ for gaps.

		A	Y	P	C	C	K	F
	0	-8	-16	-24	-32	-40	-48	-56
A	-8	5	-3	-11		-27	-35	-43
Y	-16	-3	13	5	-3	-11	-19	-27
P			5	23	15	7	-1	-9
C	-32	-19	-3	15	36		20	12
K	-40		-11	7	28	33	34	26
F	-48	-35	-19	-1	20		29	42

$$F_{i,j} = \max \begin{cases} F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - d \\ F_{i,j-1} - d \end{cases}$$

Indicative Solution for Question B2(a).

		A	Y	P	C	C	K	F
	0	-8	-16	-24	-32	-40	-48	-56
A	-8	5	-3	-11	-19	-27	-35	-43
Y	-16	-3	13	5	-3	-11	-19	-27
P	-24	-11	5	23	15	7	-1	-9
C	-32	-19	-3	15	36	28	20	12
K	-40	-27	-11	7	28	33	34	26
F	-48	-35	-19	-1	20	26	29	42

$$F_{i,j} = \max \begin{cases} F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - d \\ F_{i,j-1} - d \end{cases}$$

[10 marks]

Answers

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(b) Show the path followed by the backward algorithm for Needleman-Wunsch and show the corresponding match for the two sequences shown below.

		A	D	R	A	W	P	Q	W	E
	0	-8	-16	-24	-32	-40	-48	-56	-64	-72
D	-8	-2	0	-8	-16	-24	-32	-40	-48	-56
N	-16	-9	0	-1	-9	-17	-25	-32	-40	-48
W	-24	-17	-8	-3	-4	6	-2	-10	-17	-25
P	-32	-25	-16	-11	-4	-2	16	8	0	-8
F	-40	-33	-24	-19	-12	-3	8	12	9	1
W	-48	-41	-32	-27	-20	3	0	7	27	19

$$F_{i,j} = \max \begin{cases} F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - d \\ F_{i,j-1} - d \end{cases}$$

Match

Indicative Solution for Question B2(b).

		A	D	R	A	W	P	Q	W	E
	0	-8	-16	-24	-32	-40	-48	-56	-64	-72
D	-8	-2	0	-8	-16	-24	-32	-40	-48	-56
N	-16	-9	0	-1	-9	-17	-25	-32	-40	-48
W	-24	-17	-8	-3	-4	6	-2	-10	-17	-25
P	-32	-25	-16	-11	-4	-2	16	8	0	-8
F	-40	-33	-24	-19	-12	-3	8	12	9	1
W	-48	-41	-32	-27	-20	3	0	7	27	19

$$F_{i,j} = \max \begin{cases} F_{i-1,j-1} + s(a_i, b_j) \\ F_{i-1,j} - d \\ F_{i,j-1} - d \end{cases}$$

Match

A	D	R	A	W	P	Q	W	E
-	D	-	N	W	P	F	W	-

[10 marks]

Answers

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Question B3.

(a) Write down the ordinary differential equations (ODEs) describing the set of chemical equations

$$Y \xrightarrow{k_1=1} X + Y \qquad 2\,X \xrightarrow{k_2=1} 2\,X + Y$$
$$X \xrightarrow{k_3=1} \varnothing \qquad Y \xrightarrow{k_4=1} \varnothing.$$

Indicative Solution for Question B3(a).

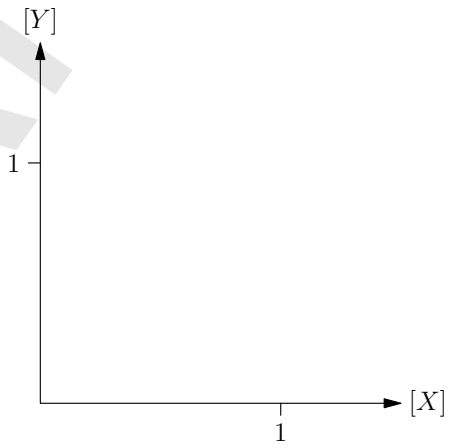
ODEs

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = [Y] - [X]$$
$$\frac{\mathrm{d}[Y]}{\mathrm{d}t} = [X]^2 - [Y]$$

Nullclines are given by $[Y] = [X]$ **and** $[Y] = [X]^2$ **with a fixed point at** $[X] = [Y] = 1$

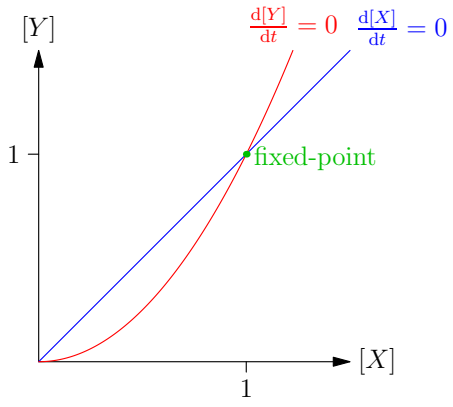
[5 marks]

(b) Sketch the nullclines on the axis shown and mark the position of the second fixed point other than $[X] = [Y] = 0$.



Indicative Solution for Question B3(b).

Answers



[5 marks]

(c) Compute the Jacobian at the fixed point.Using the fact that the eigenvalues of a 2×2 matrix,

$$\mathbf{J} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}$$

are equal to $\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$ where $T = \operatorname{tr} \mathbf{J} = J_{11} + J_{22}$ and $D = \det(\mathbf{J}) = J_{11}\, J_{22} - J_{12}\, J_{21}$ compute the eigenvalues of **J** and comment on the stability of the fixed point.

Indicative Solution for Question B3(c).

$$\mathbf{J} = \begin{pmatrix} -1 & 1 \\ 2\,X & -1 \end{pmatrix}\Bigg|_{([X],[Y])=(1,1)} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}$$

So that $D = \det(\mathbf{J}) = -1$ and $T = \operatorname{tr} \mathbf{J} = -2$. Thus $\lambda = -1 \pm \sqrt{2}$ and the fixed point is unstable as one of the eigenvalues is greater than 0.

[10 marks]

END OF PAPER

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Answers